

Smart Blind Stick for Visually Impaired (Multi-Sensor Based System)

This project report presents the detailed design, implementation, and testing of a Smart Blind Stick developed to assist visually impaired individuals. The system uses multiple sensors integrated with an Arduino Nano to detect obstacles, water hazards, and very close objects, and provides feedback through vibration and sound.

Introduction

Mobility is one of the biggest challenges faced by visually impaired people. Traditional white canes provide basic tactile feedback but fail to warn users about obstacles at different heights, distances, or environmental hazards such as water. Advances in embedded systems and low-cost sensors make it possible to enhance traditional walking aids with intelligent features.

The Smart Blind Stick is designed to improve safety and independence by providing real-time audio and haptic feedback based on surrounding conditions.

Problem Statement

Visually impaired individuals often encounter obstacles that cannot be detected by conventional walking sticks, leading to accidents and injuries. There is a need for a low-cost, portable, and reliable assistive device that can detect obstacles and hazards before physical contact occurs.

Objectives

- To detect obstacles at different distances using an ultrasonic sensor
- To detect very close objects accurately using an IR sensor
- To detect water on the ground using a water sensor
- To provide intuitive feedback using vibration and sound

System Architecture

The system architecture consists of an Arduino Nano acting as the central controller. Sensors continuously send data to the microcontroller, which processes the inputs using a priority-based decision algorithm. Output devices provide feedback to the user.

Hardware Components Description

Arduino Nano: Acts as the brain of the system and processes sensor data.

Ultrasonic Sensor (HC-SR04): Measures the distance of obstacles ahead using ultrasonic waves.

IR Obstacle Sensor: Detects very close obstacles where ultrasonic sensors may be less effective.

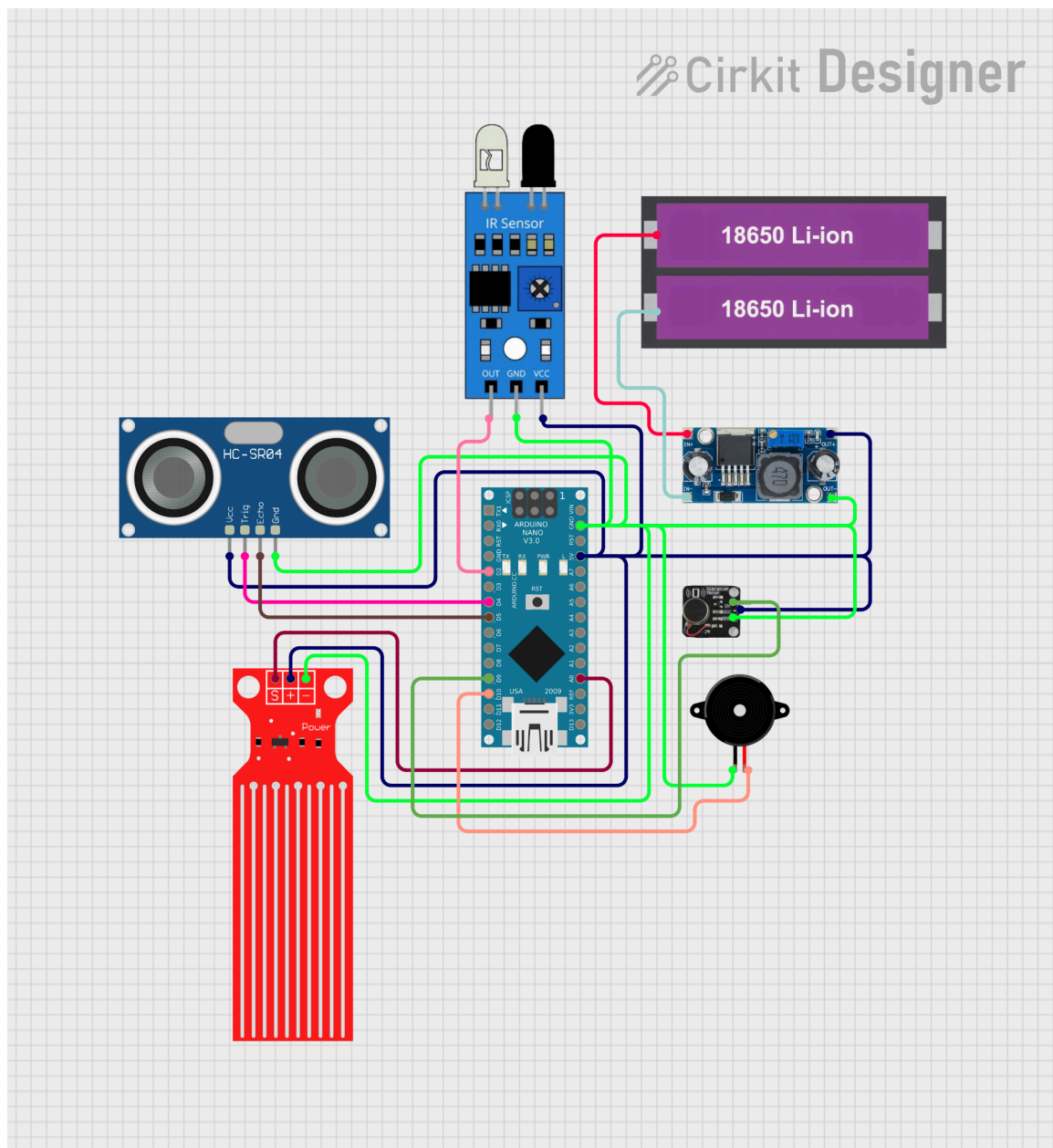
Water Sensor Module: Detects the presence of water on the ground.

Vibration Motor Module: Provides haptic feedback with varying intensity.

Passive Buzzer: Provides audio feedback with varying frequency.

Battery Power Supply: Enables portable and independent operation.

Connection Diagram:



Software Design

The software is developed using Arduino IDE and written in Arduino C/C++. The program continuously reads sensor values and applies a priority-based decision structure. Water detection has the highest priority, followed by IR-based close obstacle detection, and finally ultrasonic distance measurement.

PWM (Pulse Width Modulation) is used to control vibration intensity, while frequency-based control is used for the passive buzzer.

The code:

```
// ===== PIN DEFINITIONS =====

#define TRIG_PIN    4

#define ECHO_PIN    5

#define IR_PIN      2

#define WATER_PIN   A0

#define MOTOR_PIN   9      // PWM

#define BUZZER_PIN  10

// ===== VARIABLES =====

long duration;

int distance;

int waterValue;

int irState;

// ===== SETUP =====

void setup() {

    // --- Pin Modes ---

    pinMode(TRIG_PIN, OUTPUT);

    pinMode(ECHO_PIN, INPUT);

    pinMode(IR_PIN, INPUT_PULLUP);

    pinMode(MOTOR_PIN, OUTPUT);

    pinMode(BUZZER_PIN, OUTPUT);
```

```

// --- FORCE SAFE STATE AT BOOT ---

digitalWrite(MOTOR_PIN, LOW);    // Motor OFF

noTone(BUZZER_PIN);              // Buzzer OFF


Serial.begin(9600);

Serial.println("=====");

Serial.println(" SMART BLIND STICK SYSTEM (STABLE) ");

Serial.println("=====");
}

// ===== ULTRASONIC FUNCTION =====

int getDistance() {

    digitalWrite(TRIG_PIN, LOW);

    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);

    delayMicroseconds(10);

    digitalWrite(TRIG_PIN, LOW);

    duration = pulseIn(ECHO_PIN, HIGH, 30000); // 30ms timeout

    if (duration == 0) return 200;              // No object

    return duration * 0.034 / 2;

}

// ===== LOOP =====

void loop() {

    // --- Read Sensors ---

    distance = getDistance();

```

```

waterValue = analogRead(WATER_PIN);

irState = digitalRead(IR_PIN);

// --- SERIAL MONITOR OUTPUT ---

Serial.println("-----");

Serial.print("Ultrasonic Distance: ");

Serial.print(distance);

Serial.println(" cm");

Serial.print("Water Sensor Value: ");

Serial.println(waterValue);

Serial.print("IR Sensor State: ");

Serial.println(irState == LOW ? "OBJECT DETECTED" : "CLEAR");

// ===== PRIORITY LOGIC =====

// ① WATER DETECTION (HIGHEST PRIORITY)

if (waterValue > 100) {

    Serial.println("STATUS: WATER DETECTED !!!");

    Serial.println("ACTION: CONSTANT VIBRATION + BUZZER");

    analogWrite(MOTOR_PIN, 255);

    tone(BUZZER_PIN, 2500);

    delay(100);

    return;

}

// ② IR SENSOR (VERY CLOSE OBJECT)

```

```
if (irState == LOW) {

    Serial.println("STATUS: VERY CLOSE OBJECT (IR)");

    Serial.println("ACTION: STRONG VIBRATION + BUZZER");

    analogWrite(MOTOR_PIN, 255);

    tone(BUZZER_PIN, 2200);

    delay(100);

    return;

}

// ③ ULTRASONIC DISTANCE BASED ALERTS

if (distance > 120) {

    Serial.println("ZONE: SAFE ZONE");

    Serial.println("VIBRATION: OFF | BUZZER: OFF");

    analogWrite(MOTOR_PIN, 0);

    noTone(BUZZER_PIN);

}

else if (distance > 70 && distance <= 120) {

    Serial.println("ZONE: FAR OBJECT");

    Serial.println("VIBRATION: LOW | BUZZER: SLOW");

    analogWrite(MOTOR_PIN, 120);

    tone(BUZZER_PIN, 1000);

    delay(200);

    noTone(BUZZER_PIN);

    delay(400);

}

else if (distance > 30 && distance <= 70) {

    Serial.println("ZONE: MEDIUM DISTANCE");
```

```

Serial.println("VIBRATION: MEDIUM | BUZZER: MEDIUM");

analogWrite(MOTOR_PIN, 170);

tone(BUZZER_PIN, 1500);

delay(200);

noTone(BUZZER_PIN);

delay(200);

}

else { // distance <= 30

    Serial.println("ZONE: VERY CLOSE OBJECT (ULTRASONIC)");

    Serial.println("VIBRATION: STRONG | BUZZER: CONTINUOUS");

    analogWrite(MOTOR_PIN, 255);

    tone(BUZZER_PIN, 2200);

}

delay(100);

}

```

Working Principle

When the system is powered on, all sensors begin monitoring the environment. The ultrasonic sensor measures the distance to obstacles ahead. Depending on the distance, the system generates different vibration intensities and buzzer patterns. The IR sensor detects very close obstacles and immediately triggers strong alerts. The water sensor overrides all other inputs when water is detected to prevent slipping hazards.

Power Supply and Stability Considerations

The system is powered using a 7.4V battery, which is connected to the 5V pin of the Arduino Nano via step down DC to DC buck converter. All components share a common ground.

Advantages

- Enhances user safety and independence
- Low-cost and easily reproducible
- Lightweight
- Uses simple and readily available components

Future Scope

- Integration of GPS navigation
- Voice-based feedback system
- Smartphone application connectivity
- Improved enclosure and ergonomic design

Conclusion

The Smart Blind Stick project successfully demonstrates the application of sensor fusion and embedded systems in assistive technology. The system provides reliable obstacle and hazard detection with intuitive feedback, making it a valuable aid for visually impaired individuals.